

University of Asia Pacific

Department of Computer Science & Engineering

Laboratory Report

Chip-Design (Full Adder)

CSE 450: Design and Testing of VLSI Lab

Submitted by:

Name: Forhad Siddique Rajon

Student ID: 22201025

Section: A

Submitted to:

Md. Rasheduzzaman,

Lecturer,

Department of Computer Science and Engineering

DATE OF SUBMISSION: 25-09-2026

1. Title of the Experiment

Design and simulation of a 1-bit Full Adder built entirely from NAND gates that demonstrate the universal-gate property of NAND and implemented and verified in Xilinx ISE.

2. Objectives

- Understand the working principle of a Full Adder and its Sum/Carry Boolean equations.
- Implement AND, OR and XOR gates using only NAND gates (universal gate realization).
- Build a Half Adder and a Full Adder structurally from these NAND-based gate modules.
- Generate and verify the RTL schematic of each module in Xilinx ISE.
- Simulate the design and verify its truth table against the expected Full Adder outputs.

3. RTL Schematic Diagrams

The figures below show the RTL schematic of each module as generated in Xilinx ISE, built up from the base NAND gate to the complete Full Adder.

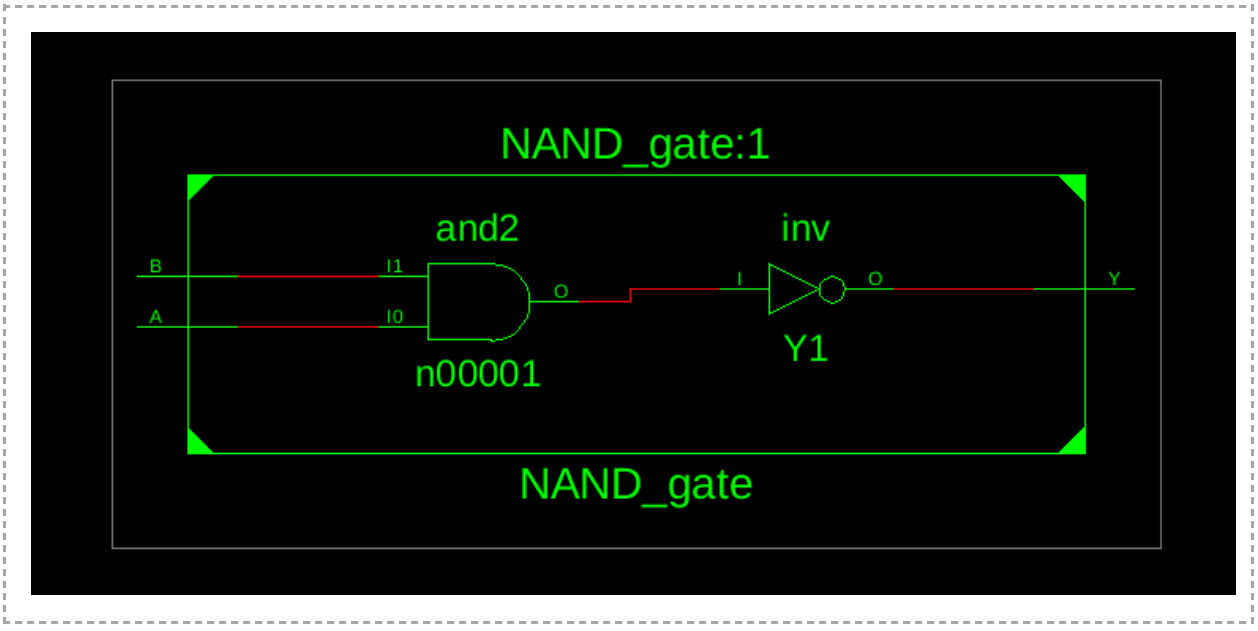


Figure 1: RTL Schematic of the NAND-gate

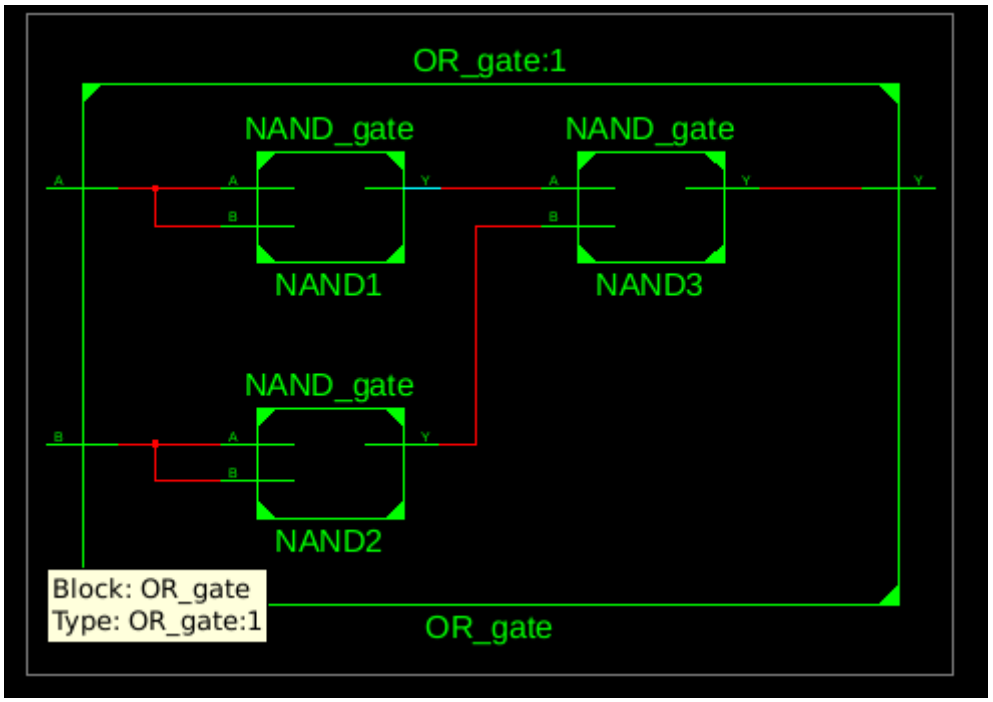


Figure 2: RTL Schematic of the OR-gate

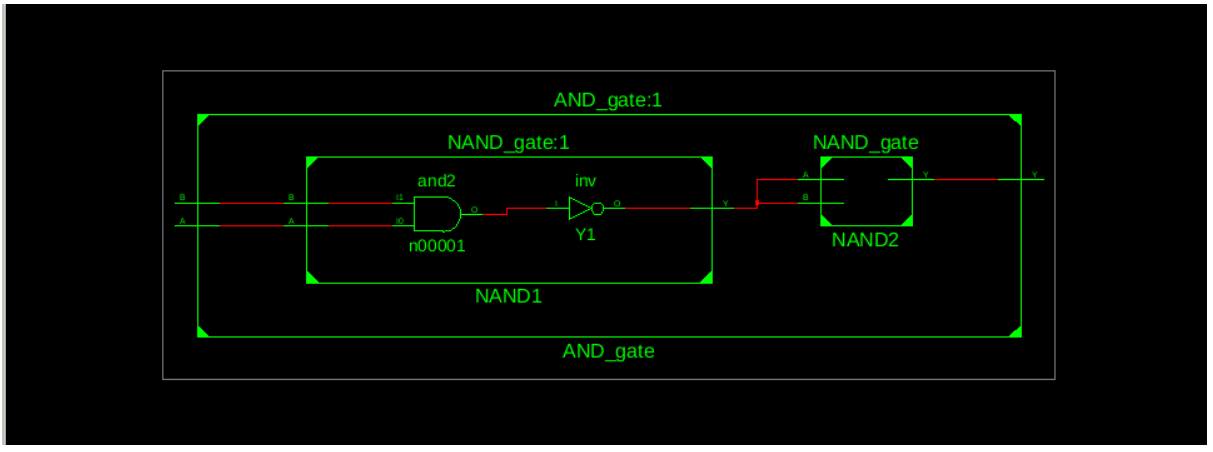


Figure 3: RTL Schematic of the AND-gate

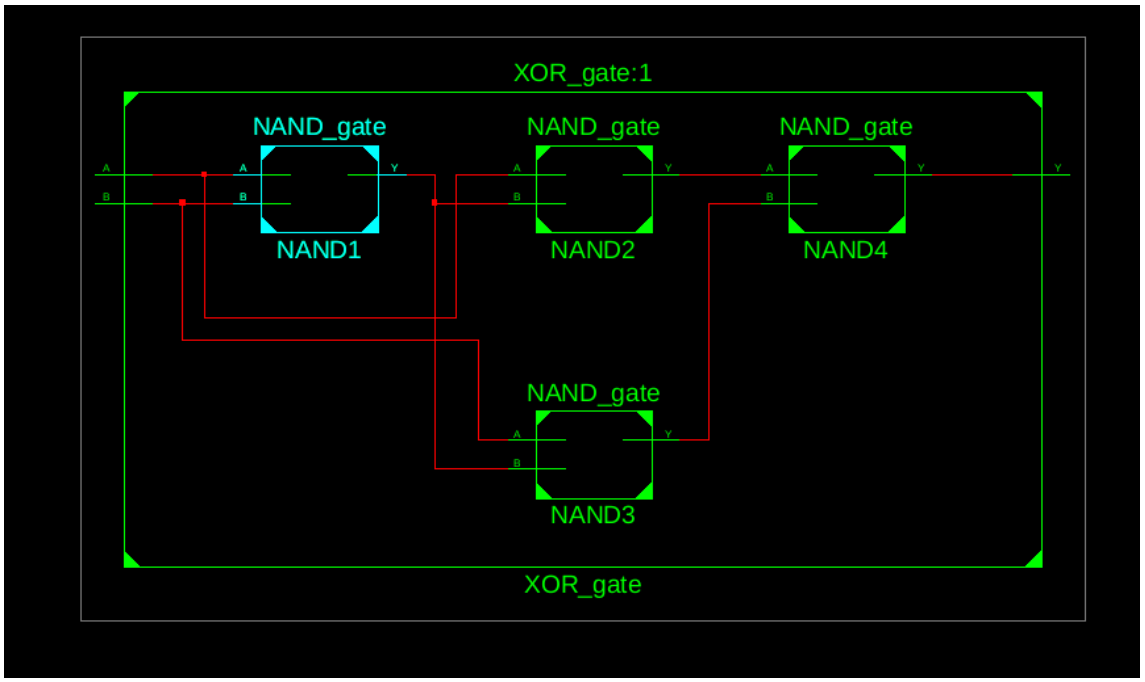


Figure 4: RTL Schematic of the XOR-gate

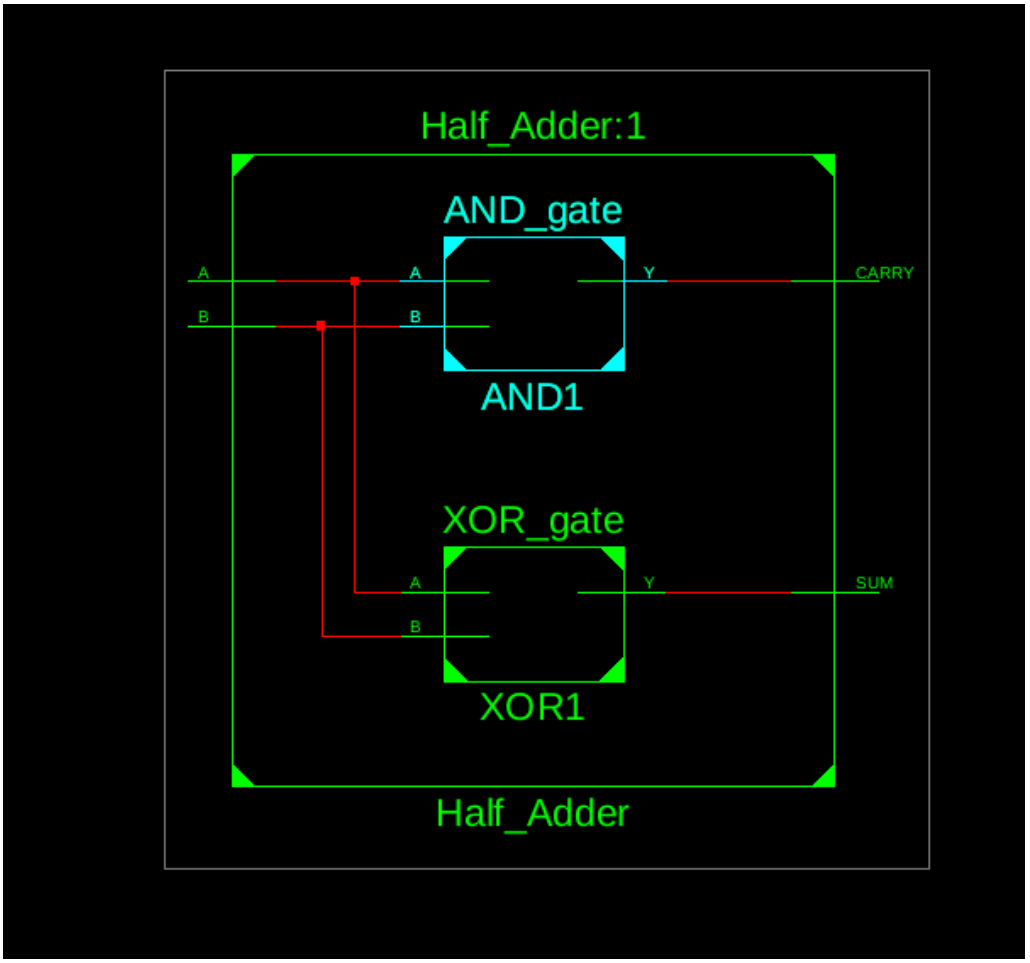


Figure 5: RTL Schematic of the Half Adder

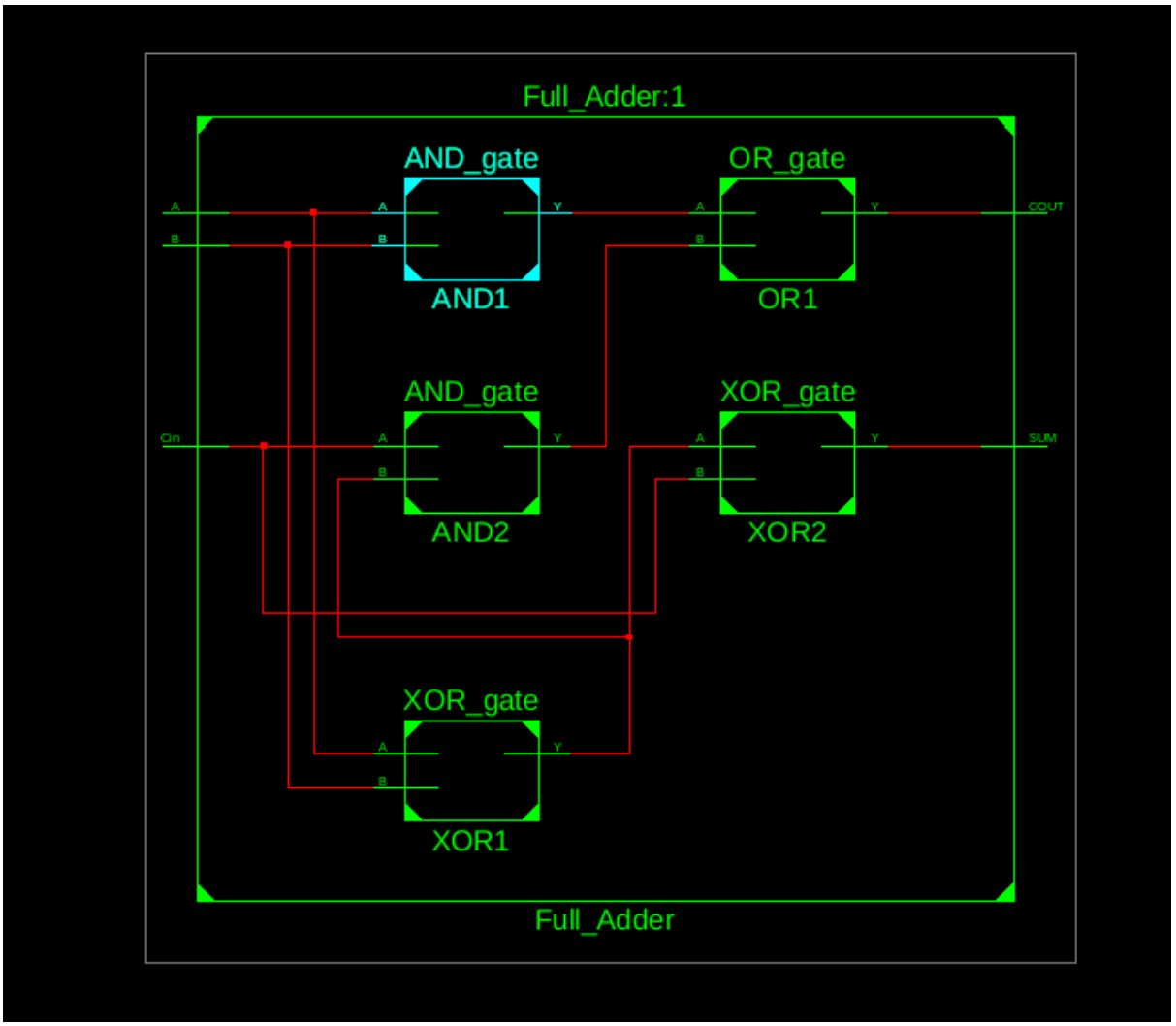


Figure 6: RTL Schematic of the Full Adder

4. Code in GitHub Repository

The complete Verilog/VHDL source code, testbench, and project files for this experiment are available in the GitHub repository below.

Repository: *Link*

Repository contents:

- nand_gate.v / nand_gate_tb.v — base NAND gate module and testbench
- and_gate.v / and_gate_tb.v — AND gate built from NAND gates, with testbench
- or_gate.v / or_gate_tb.v — OR gate built from NAND gates, with testbench
- xor_gate.v / xor_gate_tb.v — XOR gate built from NAND gates, with testbench
- half_adder.v / half_adder_tb.v — Half Adder module and testbench
- full_adder.v / full_adder_tb.v — top-level Full Adder module and testbench

5. Simulations

The design was functionally simulated in the Xilinx ISE Simulator (ISim) to verify correct operation against the Full Adder truth table.

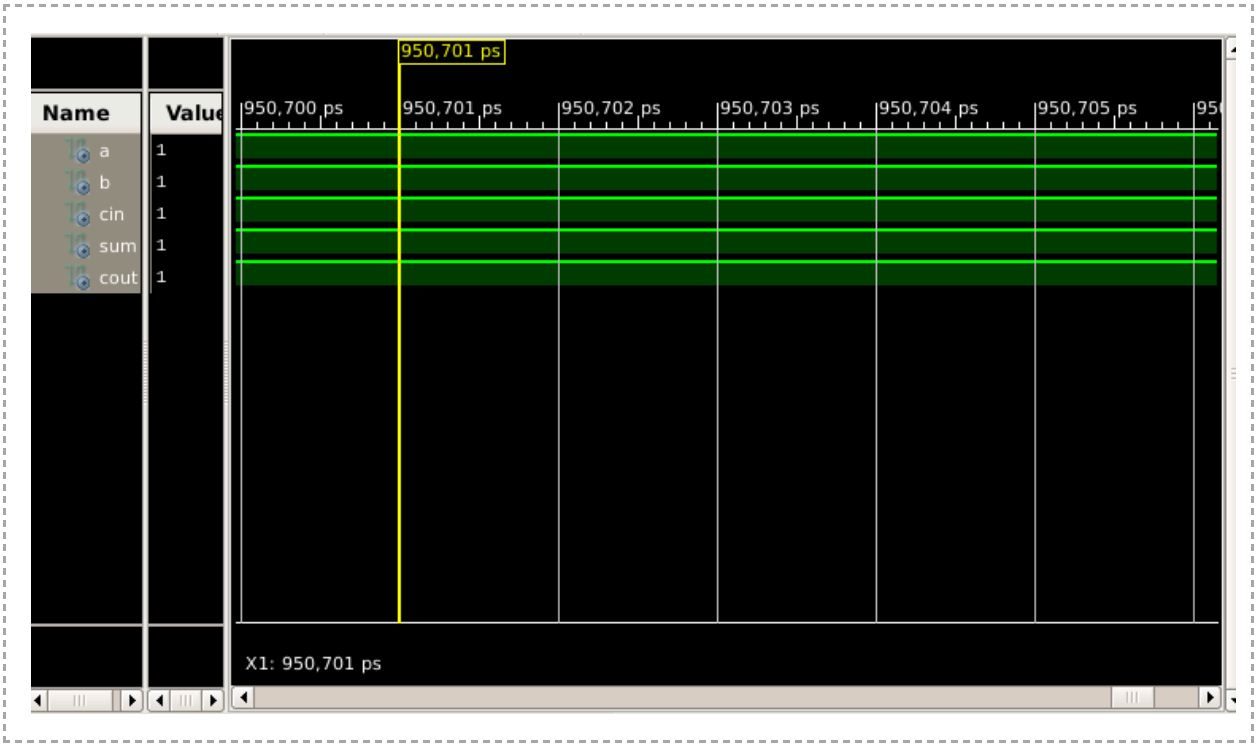


Figure 7: Design Simulation Waveform

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 1: Full Adder truth table

6. Conclusion

In this experiment, a Full Adder circuit was successfully designed using only NAND gates, demonstrating the universal property of the NAND gate. The AND, OR, and XOR functions required for the Sum and Carry logic were each realized structurally from NAND sub-modules and integrated into a complete Full Adder in Xilinx ISE. The RTL schematic confirmed the correctness of the structural design, and simulation in ISim was used to verify that the circuit's outputs matched the expected truth table, confirming that the design meets the objectives of the experiment.